Digital Game Design 🎮

Brownie Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

We’re not just coding today 💚

We’re making something **fun and fair for a real player**.
That’s what “coding for good” means in game design.

What makes a game a GAME? 🤔

Let’s take apart a game you already know — tag, or tic-tac-toe.

Every game has: - A **goal** 🎯 — what you’re trying to do - **Rules** 📜 — what you’re allowed to do - Something that makes it **fair and fun** 😄

Every video game designer starts with these same three pieces.

Designers plan FIRST 

Before we touch a screen, fill in your **game recipe**:

1. My game's **goal** is…
2. **How you win**…
3. What makes it **fun**…

Keep it **small**: one goal, one way to win.

Build: your Collector Game 🌟

Follow your mentor, one step at a time:

1. A **player sprite** that moves with the arrow keys
2. A **good thing** to collect — a star, an apple 🍎
3. A **score** that goes up when you collect it
4. A **win message** when your score hits the target

Finished early? Add a sound 🔊 or a “bad thing” to avoid!

Bugs are GOOD! 

When your game doesn’t work, that’s a **bug**.
Finding one means you’re doing **real programming**.
Fixing it is called **debugging**.

Playtest Swap 🔄

Swap seats. Play your neighbor’s game!

Then give **two stars and a wish**: - ⭐ One kind thing you liked - ⭐ One more kind thing you liked - 💫 One idea to make it even better

Playtesting is a real game-design job.

Robot moment 

Every single year, a **brand-new game** is invented for robotics competitions.

Our team designs a whole strategy around fresh rules — exactly like designing a game.

Hands up! ✋

Whose game **changed** after someone else played it?
That’s how real designers work.

You did it! 🎉

Today you: - ✅ Named the parts of a game: goal, rules, fair and fun - ✅ Planned your game on paper first - ✅ Built a game you can actually play - ✅ Made it better after a playtest

You are a game designer now!

Keep going at home! 🏠

- **MakeCode Arcade:** arcade.makecode.com (free!)
- **Scratch:** scratch.mit.edu — make games and stories
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.